A small JEPA world model: the encoder knows, the predictor does not add

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Abstract

We trained a LeWM-style JEPA world model (W1) on 182 teleoperated demonstrations of one SO-101 tabletop task for $3.30 and ran it through 20 pre-registered uses taken from published world-model papers. The encoder reads the outcome: on 69 policy rollouts it never saw in training, a linear probe on the final-frame latent separates success from failure at AUROC 0.9545, level with a hand-built colour judge (0.9432) and above a frozen DINOv2-S, the commanded joints and policy identity. The predictor adds nothing: at the grasp anchor the imagined latent reads the grasp worse than the current one (0.70 vs 0.82), and imagining forward does not add to what the current latent already reads. Every use that needs a multi-step rollout (ranking, surprise monitoring, risk, multi-scale planning) misses its bar. One rig, one model, one data scale; the controls are exploratory.

1. Introduction

World-model papers report the use that worked. We report every use we pre-registered for one small JEPA world model on one $200 arm: 20 gates, each with a bar, a dataset and a verdict, logged in a public register. Audited against the commit history (20 of 20 rows matched to the audited pre-registration check), 19 bars were committed before their result and 1 (G6b) in the same commit as the result; one row, the on-device latency of the pretrained RoboJEPA / V-JEPA 2.1, is not a W1 use. Of the 20, 3 pass, 1 is met as predicted, 2 are not proven and 14 fail or are no-go (the RoboJEPA row is one of the 14). The result splits along the model. The *encoder* reads the episode's outcome from the current frame, better than a frozen foundation encoder or the robot's own joint commands, including at the grasp anchor, 15 frames before the gripper first closes. The *predictor*, which every forecasting use relies on, does not improve on that by imagining forward. Contribution: (i) the register, with a load audit of every checkpoint use (Fig. 1); (ii) controls that separate encoder from predictor (Table 2, Fig. 2); (iii) the reading that, at this data scale, the predictor is the weak link. **Caveats stated up front.** The encoder readouts are linear probes fit with leave-one-session-out (LOSO) cross-validation on 69 rollouts (44 success / 25 failure, 13 sessions) with operator labels; W1 was trained on teleop only and never saw them, but the controls were chosen after the data and are **exploratory**. 3 of the 3 passes have bars committed before the result; G10's is retrospective. Ordered by commit alone, without the audit, the register reads 13 bar-first and 6 same-commit.

2. Setup

Robot and data. One SO-101 arm, two cameras (top, wrist), one task family (put a bowl in another). W1 is trained on `ravediamond/so101_bowls_all_trimmed` (157 train / 25 validation episodes, split by grid layout); the evaluation rollouts are `ravediamond/so101-bowls-policy-rollouts`, a separate dataset of SmolVLA rollouts recorded after W1 was trained. **Model.** W1 follows the LeWM recipe (multi-view ViT encoder, action-conditioned predictor, SIGReg), trained on one rented GPU. **Controls.** W2′: frozen DINOv2-S, mean-pooled, top + wrist. Joints: the commanded joint vector. G9c-3: a hand-built colour judge on the final frame (no learning). Policy identity: which checkpoint ran the episode. **Load audit.** One early gate (G11-A) loaded W1 with `strict=False` and ran a random encoder. We therefore re-traced the load path of every use: 26 trained, 1 random (G11-A, already voided), 0 unknown. The VLA-JEPA world-model block loads 351 / 351 keys (0 missing).

3. The register

Figure 1 lists the 20 uses with what each takes from W1: the encoder alone (7), the predictor (8) or both (5). *Encoder reads.* G10 step 1 reads the outcome at rest (pass, retrospective); W1-C1's current-latent arm reads the grasp at the anchor (0.82). The pose probes (LeJEPA) measure what else the latent holds: the encoder holds about half of the joint variance linearly (R^2 0.565; the frozen DINOv2-S 0.867). Encoder uses that read progress or stage rather than outcome fail: G6a, G6b, W1-B, and SURVEYOR's latent-distance scale (τ 1.2488, paper 0.20).

use	takes from W1	bar	verdict
G0	predictor	bar-first	**pass**
C1 probe	both	bar-first	not proven
G1b W1	predictor	bar-first	fail
G1 probes	encoder	bar-first	**W2 pass**
G1 W2 dynamics	predictor	bar-first	fail
G6a	encoder	bar-first	T1 fail, T2 sanity
G10	both	bar-first	**pass (retro.)**
G6b	encoder	same-commit	fail
W1-A	predictor	bar-first	fail
W1-B	encoder	bar-first	fail
G11-B-cal	predictor	bar-first	fail
W1-C1	both	bar-first	fail
W1-C2	predictor	bar-first	fail
W1-R	predictor	bar-first	fail
REAP v1 / v2	both	bar-first	fail
§30 inverse dynamics	encoder	bar-first	not proven
LeJEPA	encoder	bar-first	**B1–B3 met**
SURVEYOR	encoder	bar-first	fail
HWM	predictor	bar-first	H1, H2 fail
RoboJEPA / V-JEPA 2.1	both	bar-first (not W1)	no-go

Figure 1: The register: every pre-registered use of W1 (and the W2 / RoboJEPA rows beside it), what it takes from the model, whether its bar was committed before the result, and the verdict. Source: repro/register/expected/FEASIBILITY.md §2, uses.json and prereg_audited.json (audited class).

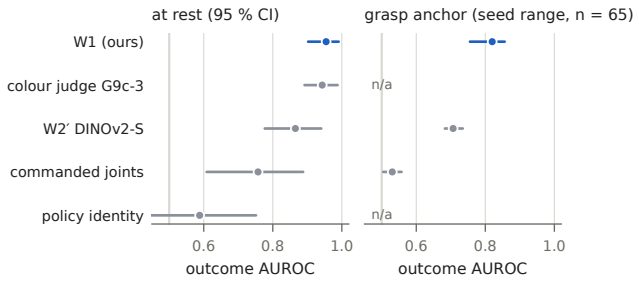


Figure 2: Table 2 as a dot plot. Left: AUROC at rest with its 95 % bootstrap CI. Right: AUROC at the grasp anchor, mean of 5 seeds with the seed range (not a CI). Exploratory, LOSO, operator labels.

Predictor forecasts. G0's open-loop prediction beats copy-last (0.31 vs 0.89 at K = 30): the predictor predicts its own latent. Every use of that prediction fails: G1b ranks chunks at top-1 0.60 (bar 0.8) while the commanded actions alone reach 0.68–0.90; W1-A's surprise catches 2/8 at 2/42; W1-C1's imagined latent (0.70) is below the current one; HWM finds no crossover; W1-C2, W1-R, REAP and G11-B-cal all miss their bars.

4. What the encoder knows

Sensitivity of the anchor column to the seed aggregation (exploratory): the AUROC of the 5-seed-averaged predictions is 0.8377 / 0.7175 / 0.5346 (W1 / W2′ / joints), and seed 0 alone gives 0.7554 / 0.7024 / 0.5574, a smaller W1-over-W2′ gap; the ordering holds in both. At rest W1 is level with the colour judge (CIs overlap) and above the frozen encoder and the joints; at the grasp anchor, where no judge applies, it leads both controls. The pose readers hold the joints better (§3) but the outcome worse: the encoder holds outcome information they do not. The original G10 gate scored W1 at 1.000, but against the G9c judge's labels, not the operator's, on 87 rollouts; the rerun here, on the 69 Orin rollouts against operator labels with the controls beside it, gives 0.9545.

5. Where the predictor fails

The cleanest contrast is W1-C1: same encoder, same anchor, current latent 0.82 vs imagined 0.70 (bar ≥ 0.85, $\geq +0.05$ over current). Imagining forward loses outcome information the encoder had. On SURVEYOR's scale check, W2′ (τ 0.3601) and the joints (τ 0.0265) pass where W1 fails, which points at W1's latent geometry rather than the task. *Exploratory, three further lines.* (1) On the distillation probe's §30 term (pooled, 25 teleop validation episodes), trained W1 (0.7476) is not distinguishable from its own random init (0.7974): difference −0.0499, 95 % paired episode-bootstrap CI [−0.7403, +0.6559] (`gap5_pooled_rerun.json`). The term is sensitive to BLAS / ridge conditioning: the same untrained network scored 1.4383 on one machine and 0.7974 / 0.8042 on two others. (2) The surprise scorer that gives W1 a class-5 AUROC of 0.775 (below its sanity bar) reproduces G0's K = 30 corruption ratios (cross-episode 3.428 vs 3.341, constant-first 2.558 vs 2.543; `M1_diag_g0_repro.json`), so 0.775 is a real value, not a load or scorer bug. (3) VLA-JEPA's

feature	at rest	grasp anchor (65 scored, 21 fail)
W1 (ours)	0.9545 [0.9023, 0.9908]	0.8199 [0.7141, 0.9041]
colour judge G9c-3 (no learning)	0.9432 [0.8919, 0.9878]	—
W2′ frozen DINOv2-S	0.8655 [0.7767, 0.9405]	0.7067 [0.5816, 0.8269]
commanded joints	0.7573 [0.6084, 0.8880]	0.5305 [0.3709, 0.6855]
policy identity	0.5882 [0.4084, 0.7515]	—

Table 1: Outcome AUROC on the same 69 rollouts. Exploratory (controls chosen after the data), LOSO linear probes, operator labels. At rest: 95 % episode-bootstrap CI. At the anchor: 95 % episode-bootstrap CI on the mean over 5 seeds of the out-of-fold AUROC (episodes resampled jointly across the seeds, 2,000 resamples, percentile).

predictor is near action-blind (exploratory, 5 anchors; `M3_diag_action_shuffle.json`): action tokens from another episode change the tokens by 21–28 % (relative L2) and the loss by about 0.05 %; zeroing the tokens changes it by about 1.05 % on average ($|\Delta|$ 0.4–2.2 %); the forward is deterministic. Its class-5 AUROC of 0.501 (below the same sanity bar) is therefore a real value, and its table stays unread under the frozen bar. *Objection: G10 step 2 is a predictive pass.* The imagined latent beat policy identity by +0.179 — but that bar is retrospective, and the two bar-first reruns of the same idea (W1-C1, G11-B-cal) fail.

6. Related work

LeWorldModel (2603.19312) trains the JEPA recipe we use end to end from pixels and plans in Push-T and simulation; its planning use fails here (G1b, HWM). LeJEPA identifiability (2605.26379) predicts SIGReg latents identify state up to anisotropy, which holds here partially. DINO-WM (2411.04983) plans on frozen DINOv2 patches, our W2 / W2′ design. V-JEPA 2.1 (2603.14482) is too slow on our Orin. ContactGuard (2608.13438) finds imagined-latent probes beat current-latent ones on grasp tasks; we find the opposite (W1-C1). ReAP (2610.16309) predicts action-conditioned risk; its re-test misses its bar here. Two results agree with ours from the control side: SaiVLA-JEPA (2610.38508) finds a training-only future-latent head adds +0.03 points to control on 10,030 paired episodes, and HIJP (2610.23615) finds LeWM below goal-conditioned behaviour cloning in simulation.

7. Limitations

Exploratory controls: 69 rollouts, operator labels, probes chosen after the data; fresh rollouts scored under a bar committed first are the test. One rig, one model, one data scale: a second scale point (653 episodes) is costed and queued separately, and the scale curve belongs to the longer version. Not run: W1-A surprise and W1-C1 imagined with W2′ / joints (each needs its own predictor), G6a T1 with the controls (needs a reimplementation). G10's bar is retrospective; the one same-commit bar is G6b's.

8. Conclusion

On a small teleop dataset, a JEPA world model learns an encoder that reads how the episode went, a short horizon ahead included, and a predictor whose imagination adds nothing to it. Reporting every pre-registered use, not the best one, is what makes that visible.

anchor (episode / frame)	true loss	Δ shuffled (%)	token change (%)	Δ zeroed (%)
8 / 402	1.3031	+0.014	21.1	+2.22
10 / 227	1.4641	+0.088	27.6	+0.50
23 / 327	1.4215	+0.036	21.1	+0.87
59 / 302	1.4637	+0.030	25.5	+1.28
65 / 302	1.5236	−0.070	22.0	−0.38

Table 2: VLA-JEPA world-model loss at 5 anchors with the true action tokens, tokens from another episode (Δ shuffled) and zeroed tokens; token change is the relative L2 of the shuffled tokens. Exploratory, a spot check, not a power analysis. Source: repro/gaps/expected/M3_diag_action_shuffle.json.